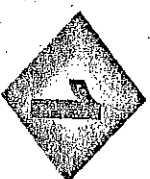


INTRODUCTION TO MICROECONOMICS



THE MEANING OF ECONOMICS

The word economics comes from a Greek word "Oikonomia" meaning the study of efficient household management. The resources of the Greek household were scarce but their material desires knew no bounds. As one need was satisfied, another one emerged, the term "Oikonomia" was applied by the Greeks to this art of rationing the severely limited resources for the satisfaction of their desires.

With the evolution of time, and continuous application of scientific reasoning to everyday living, the concept of efficient use of limited household resources has been transmitted to the entire fabric of the society. Business entities, household units, states and national governments have all become participants in the art of economising. For this broader setting, Economics analyses how a society chooses to produce, distribute and consume goods and services in a world of scarce resources.

INSTITUTIONS OF MICROECONOMICS

The economy represents a collection of institutions of each, which directly participates in the art of economising. These include individual households, business firms, organised unions and government.

Households

The household is an economic concept that refers to groups of people or persons functioning as an economic unit. It can consist of a single person, a married couple and their siblings or an extended family. Whatever its composition, each household has a source of income and a responsibility for its disposal. It is basically a consuming unit and can be used to estimate measure or predict the quantitative economic consequences of its consuming behaviour.

As an economising institution the objective function of the household is utility maximisation which depends upon current as well as future consumption, leisure etc. The instruments used in accomplishing these are the consumption levels of all goods and services, savings and occupational choice.

Introduction to Microeconomics

The limiting factor is the budget constraint. In this respect, the total expenditure on goods and services cannot exceed nominal income. The normative rule of the household is to allocate income among goods and services so that the ratio of marginal utility to price is the same for all goods and services. The household then saves an amount, which depends upon current income, expected future income and the utility of present and future consumption and future prices.

Firms

The firm is a business concern usually involved in production. It could be a sole proprietorship, a partnership or a corporation and may produce one or more products. A firm belongs to a specific industry but may consist of one or more plants. The business of the firm involves the production of goods and services for sale to other firms or to final consumers.

As an economising institution, its objective function could be output maximisation, profit maximisation, maximisation of sales, cost minimisation, but the ultimate objective of the firm is profit maximisation. The basic instruments are output levels and factor inputs. The constraints are technology and capital requirements. Output depends on the production function. The normative rule of the firm is to equate the marginal revenue products to factor prices for all factor inputs.

Trade Unions

Trade unions are associations of employees organised to initiate and win, through collective bargaining, wage increases to match productivity gains.

Government

A government is a political entity with important macro-economic functions affecting the entire economy. As an economising unit, the objective function of the government is the maximisation of the "social welfare function", which depends upon the level of employment, production, purchasing power of money, economic growth, cyclical instability, inequities in distribution etc. The basic instruments are monetary and fiscal policy, debt management, pricing and regulatory policy. The normative rule involves the use of automatic stabilisers, which automatically offsets undesirable changes.

THE THEORETICAL FOUNDATION OF MICRO-ECONOMICS

By the eighteenth century the philosophy of individualism had gained grounds in Europe. The philosophy was developed in parts and nursed over the years by men like Prince Machiavelli, David Hume, John Locke, and Hegel to name a few. It was Adam Smith (1723-1790), however, who refashioned and welded it solidly into its proper place in:

Introduction to Microeconomics

For Michael Smith

economic theory. In its brewing stages, individualism struck fears into the minds of social philosophers and the vital question of "how to ensure social order out of the potential chaos of an individualistic society became an intellectual preoccupation of the scholars of the time. It was out of the intellectual response of social philosophers to this question that the classical theory of democracy emerged. This theory maintained essentially that men are inherently selfish and would institute governments to protect their natural rights as individuals. John Locke (1632-1704), argued that order and economic freedom were compatible and governments were instituted to avoid chaos and preserve natural rights, but absolute power be granted to no one (Fusfeld, Op. Cit, P. 21). But the question of how interdependent economic units, and selfish ones at that, could operate harmoniously without holding the society to ransom was yet to be answered.

The solution was economic freedom proposed by Adam Smith. In his book, "The Wealth of Nations" (1776), the classic masterpiece which established Smith as the father of modern economics, he recommended a system of natural liberty, in which each individual is left free to pursue and advance his own interest. This system, Smith contended, would guarantee the greatest wealth for both the individual and the society.

In his analysis, Smith maintained that the driving force which guides man to whatever work the society is willing to pay for is self-interest. "It is not from the benevolence of the butcher, the brewer or the baker that we expect our dinner, but from their self interest. We address ourselves not to their humanity but to their self love, and never talk to them about our necessities but of their advantage". (Ibid).

However, self-interest only propels man into action. Something else must prevent the cravings of profit-hungry individuals from staking up the society. According to Smith, a society activated by self-interest alone will be a society of ruthless profiteers. This strict beneficial consequences of conflicting self-interest. Thus anyone who allowed himself to be misguided by self-interest to exceed the common denominator of acceptable behaviour will find his trade quickly over taken by similarly motivated individuals who are in the same trade. Smith's immutable law of the market demonstrates how the drive of individual self-interest in an environment of similarly motivated individuals will result in competition, guaranteeing the provision of those goods that the society wants, in quantities and qualities the society desires and at prices they can afford to pay.

This same market mechanism controls the allocation of the productive resources of the society to meet its changing desires without any decree or a dictum or output schedules from a planning authority.

Just as the market mechanism regulates prices, qualities and quantities of goods and services produced, it also regulates the incomes of those who cooperate to produce them.

The wages and salaries of the workers, the rent for the owners of land, the interests on capital and the profits of the entrepreneur are all determined by the same task-master, the market mechanism. The system of economic freedom (*laissez-faire*), guided by Smith's immutable law of the market was a beautiful, self functioning, self adjusting, self perpetuating mechanism. If output levels, prices or remuneration should swing off the socially ordained level, market forces were set in motion to lead them back to order. You may do as you please in the market. But to the anonymous dispensation of the impersonal market mechanism there can be no appeal.

Thus, classical microeconomics assumes away government intervention in the working of the economic system. The self regulating market mechanism will always ensure that the economy is at the full employment level. Factor inputs available for work at the determined prices will always be fully employed and, thus, society's output will always be at the full – employment level. Any deviation from the full employment output level is automatically corrected through adjustments in prices (prices of factor inputs and prices of output).

The self-functioning, self-adjusting and self-perpetuating market mechanism of the classical economists is illustrated in the simple circular flow of income and output.

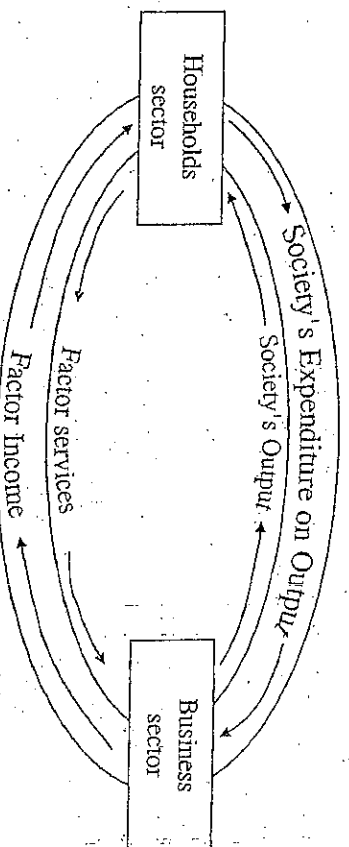


Fig 1.1: The simple circular flow of output, income and expenditure

THE FUNDAMENTALS OF SUPPLY AND DEMAND

Much of the wisdom of the economics profession, contributed over the years by men like Smith, Richardo, Marshall, Keynes, Samuelson and Friedman, purporting to explain how the economy works, all anchor on the simple concept of supply and demand. The concepts of supply and demand are analytic tools generally employed by economists to describe the way markets function.

The classical economists, Smith, Richardo and others, thought that the price of a commodity was determined by its cost of production. The *marginalists* emphasised the role of demand and consumer preferences in determining prices. It was Alfred Marshall

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(1842-1924), the father of modern microeconomics, who unified these two propositions by introducing the twin concepts of supply and demand. Marshall proposed the laws of supply and demand, according to which the quantity demanded of a commodity was inversely related to price and supply of a commodity was largely influenced by the cost of production. In conclusion Marshall proposed that the market price of a commodity was determined by the balancing of the two forces of supply and demand. On the relative strength of the two forces as determinants of market prices, Marshall pointed out that in the short-run, both supply and demand were equally important in determining prices. However, in the long-run the cost of production which largely influenced supply was the crucial determinant of market price for commodities.

THE BASIC NATURE OF A MARKET

The term market generally refers to economic institutions for the commercialisation of goods and services. It does not basically imply a location where goods and services are physically exchanged. The term also is applied to the summation of all transactions (purchases and sales) involving a specific good or service.

It is difficult to specify a particular market precisely. This is largely due to the difficulty economists encounter in defining the commodities to be classified under a particular market. We may talk of the grains market, the oil market, the automobile market and so on. In some casual sense all these markets do exist. However, in more specific terms it is difficult to identify a particular market for a specific product because all markets seem to merge at one end of the spectrum and split into numerous pieces at the other end. Thus we can speak of the grains market just as we can also speak of the corn or wheat market. Clearly, the degree of substitutability is important in determining whether two products are in the same market or not. Thus as one group's products that are distant substitutes in the same market, the concept of the market becomes less precise.

The motivating forces that make the market mechanism to work are the conflicting interests of selfish buyers and sellers and competition. Conflict stems from the fact that the two basic sides of the market have conflicting interests. Suppliers want to sell at the highest possible prices. Buyers on the other hand want to purchase at the lowest possible price. In a market, all participants want to buy cheap and sell dear. As the sellers push for the highest possible price and buyers lobby for the lowest, the resulting conflict between the basic sides of the market is resolved by competition.

In general, prices tend to rise when goods become scarce, and when they are in abundance, prices tend to fall because competition will bid the price down. The twin forces of competition and self-interest make the market to operate effectively. Sellers will always strive to obtain the highest possible prices for their products, bearing in mind that if the price is too high competitors will make all the sales. Buyers lobby for the lowest

possible price while noting that goods will in the end go to the highest bidder. The prices in most markets are determined in this manner by these impersonal market forces.

There are always two sides to a market, namely the supply side and the demand side.

Many of the subsequent chapters will analyse in greater details the supply and demand aspects of the market. This chapter focuses on the very important fundamentals of supply and demand, including the determinants of equilibrium price and quantity in the market.

THE DEMAND SIDE OF THE MARKET

At any point in time, the quantity of any product that consumers will purchase depends on a number of key variables namely, the price of the products, the prices of the substitutes and complements, income or wealth of consumers, tastes and preferences of consumers and consumers expectations concerning future prices and incomes. All these variables play a role in what is called the demand function for a particular good.² More relevant variables influences the quantity of a product that consumers will buy per unit of time. A demand function is conventionally represented by an equation of the form.

$$Q_d = f(P_x, I, P_o, T, N, E) \quad (1.1)$$

Where

Q_d = The quantity of good x demanded per unit time

P_x = The price per unit of the good

I = Income, wealth or other measures of purchasing power of consumers.

P_o = The prices of other commodities (substitutes and complements)

T = Tastes and preferences

N = Population

E = Expectation regarding future market conditions.

The magnitude of the impact of changes in each of the demand function variables on the quantity demanded may vary from one product to another. In the demand function for items of daily consumption, such as *bourvita*, for example, changes in the price of substitutes may be significant. In the demand function of consumer durables such as motor cars, changes in income, changes in credit terms etc may be more important.

Generally, the term demand is used to capture the relationship between the set of possible prices of a product and the quantities of the product that consumers are willing and able to buy at such prices per unit of time, other things being equal. This relationship can be expressed in an equation form as,

$$Q_d = f(P_x, I, P_o, T, N, E) \quad (1.2)$$

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This means that the demand for the commodity (x) is a function of its own price, all other demand function variables held constant. The vertical bar between P_x and other demand function variables is interpreted to mean that these variables are held constant.

The Demand Schedule and the Law of Demand

A *demand schedule* is a table showing the relationship between various possible prices of a commodity and the quantity purchased per unit time. For a *normal good*, this relationship between prices and quantities demanded will be inverse (leading to a downward sloping demand curve). This inverse relationship is known as the *law of demand*. In general, the law of demand states that over a specified period of time, consumers will tend to buy more of a commodity when the price is low than when it is high.

There are three major reasons, which pointedly attest to the *validity of the law of demand*. First, there is a general tendency that any rational consumer will substitute relatively cheaper products for those that are relatively more expensive. This tendency is called the *substitution effect*. Given two goods X and Y , which are close substitutes. If the price of X is increased while the price Y remains constant, the substitution effect dictates that the quantity demanded of Y will increase while the demand for X will fall, and vice versa. We can express this relationship symbolically as,

$$\begin{array}{l} P_x \uparrow \bar{P}_y: Q_x \downarrow \rightarrow Q_y \uparrow \\ \text{Thus } P_x \uparrow \rightarrow Q_x \downarrow \end{array}$$

A price change also generates an *income effect*. When the price of a particular product increases while consumers income remains unchanged, then the same amount of income can now command fewer units of that commodity. Thus, as price increases, purchasing power or ability to buy falls and quantity demanded falls.

Thirdly, as a consumer acquires additional units of a particular product, there is a tendency for him to be saturated with the product, his total satisfaction (utility) increases but the additional (marginal) utility accompanying each additional unit of the product becomes less and less. This tendency is called the *law of diminishing marginal utility*. Since every rational consumer will always equate the price he pays for any product with the marginal utility of that product, as the quantity he consumes of a commodity increases, the price the consumer is willing to pay for an additional unit falls.

These three factors (the substitution effect, the income effect and the law of diminishing marginal utility) operate in concert to maintain and perpetuate the fundamental law of demand.

How the effect the law of fundamental law of demand.

There are however exceptions to that law. For a "giffen good", a price fall may elicit a fall in demand. While the substitution effect of price changes on the demand for a giffen good is negative (as in the case of normal goods), the income effect is positive and stronger than the substitution effect. The result is a net positive relationship between demand and price. In general, giffen goods tend to have abnormal demand curves at low prices (Fig. 1.2a). The demand for 'ostentatious goods' also exhibit a positive relationship between price and quantity demanded especially at relatively high prices (Fig 1.2c)

Since the demand function is a relation between quantity demanded and price, changes in price will lead to movement along the same demand curve while changes in one or more of the other determinants of demand will lead to shifts in the demand curve. Thus, other factors (outside of price) which influence the demand functions are called *shift factors*.

Types of Demand Functions and Curves

Given that the quantity purchased of a commodity is a function of its price, other determinants remaining constant, then we can represent the demand function as,

$$Q_d = f(P) \quad (1.3)$$

$$\frac{dQ_d}{dP} = f_p < 0, \text{ for normal goods}$$

Sometimes, a demand function may be expressed as,

$$P = f(Q_d) \quad (1.4)$$

$$\frac{dP}{dQ_d} = f_q < 0, \text{ for normal goods}$$

Equation 1.4 is simply the inverse of 1.3.

A demand function may be linear or non-linear. The standard form of a linear demand function is

$$Q_d = b_0 - b_1 P$$

Where b_0 = intercept of function
 b_1 = slope of function

$$(1.5)$$

A demand function can also be non-linear (curvilinear). There are various types of non-linear demand curves. The most common is the "constant-elasticity-demand curve".

Shift factors

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The graph of this type of demand function is given in Fig. 1.2(b). The curve yields a rectangular hyperbola characterised by constant elasticity of demand at all points.

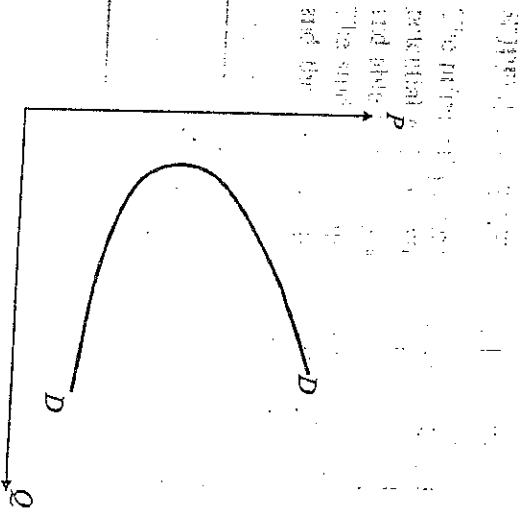
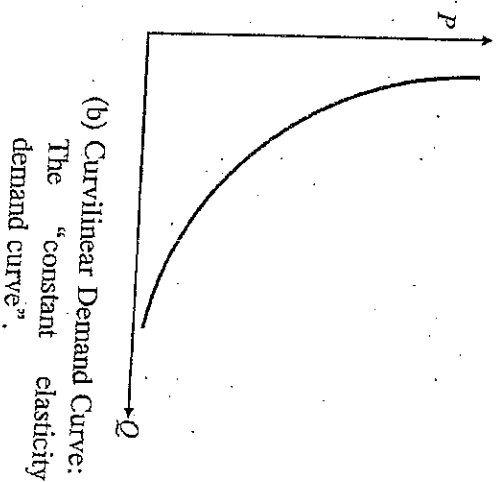
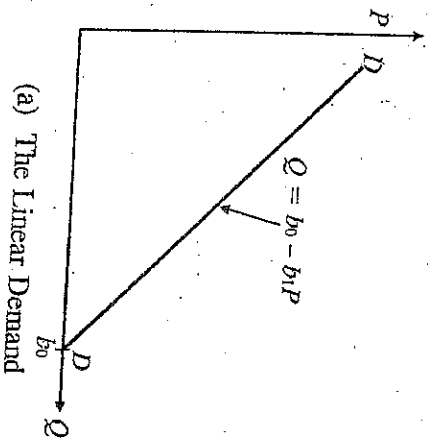


Fig 1.2: Types of Demand Curves

SUPPLY SIDE OF THE MARKET

The principal participants in the supply side of the market are the producers, sellers and potential sellers. The term supply refers to the quantity of goods that firms are willing and able to provide at various levels of prices per unit of time, other things being equal. The study of the supply side of the market involves the study of the cost of production and the objective function of the firm. The quantity that firms are willing to offer

What are the goods
normal demand supply
supply curve
29

Types of Supply Functions and Curves

Given that supply is a function of price, other determinants remaining constant, we can represent the supply function as,

$$\begin{aligned} Q_s &= f(P) \\ \text{Supply function} \quad \frac{dQ_s}{dP} &= f'_p > 0 \end{aligned} \quad (1.8)$$

The supply function can also be expressed as,

$$\begin{aligned} P &= f(Q_s) \\ \frac{dP}{dQ_s} &> 0 \end{aligned} \quad (1.9)$$

Like the demand function, a supply function may be linear or non-linear. The general form of a linear supply function is,

$$\begin{aligned} Q_s &= a_0 + a_1 P \\ a_0 &< 0 \end{aligned} \quad \begin{aligned} \text{Where } a_0 &= \text{intercept of the function} \\ a_1 &= \text{slope} \end{aligned} \quad (1.10)$$

Equation 1.11 yields an upward sloping straight line graph with an intercept a_0 and slope defined by a_1 (Fig. 1.3a). The curve, like the linear demand curve has a constant slope but different elasticity at different prices.

Supply functions may not, however, be linear. Like demand functions, supply functions can be curvilinear as is the case of the labour supply function (Fig. 1.3d).

Exceptions to the law of supply occurs,

- (i) in the very short run when quantity supplied cannot be changed at all. This situation will generate a vertical supply curve.
- (ii) when the firm must sell all the stock of goods at its disposal, e.g. in the case of perishable products. This condition *may* give rise to a downward sloping curve (Fig. 1.3c) and,
- (iii) in the case of the individual's supply of labour, when the marginal disutility of work exceeds the wage rate. This situation will yield a backward – bending labour supply curve (Fig. 1.3d).

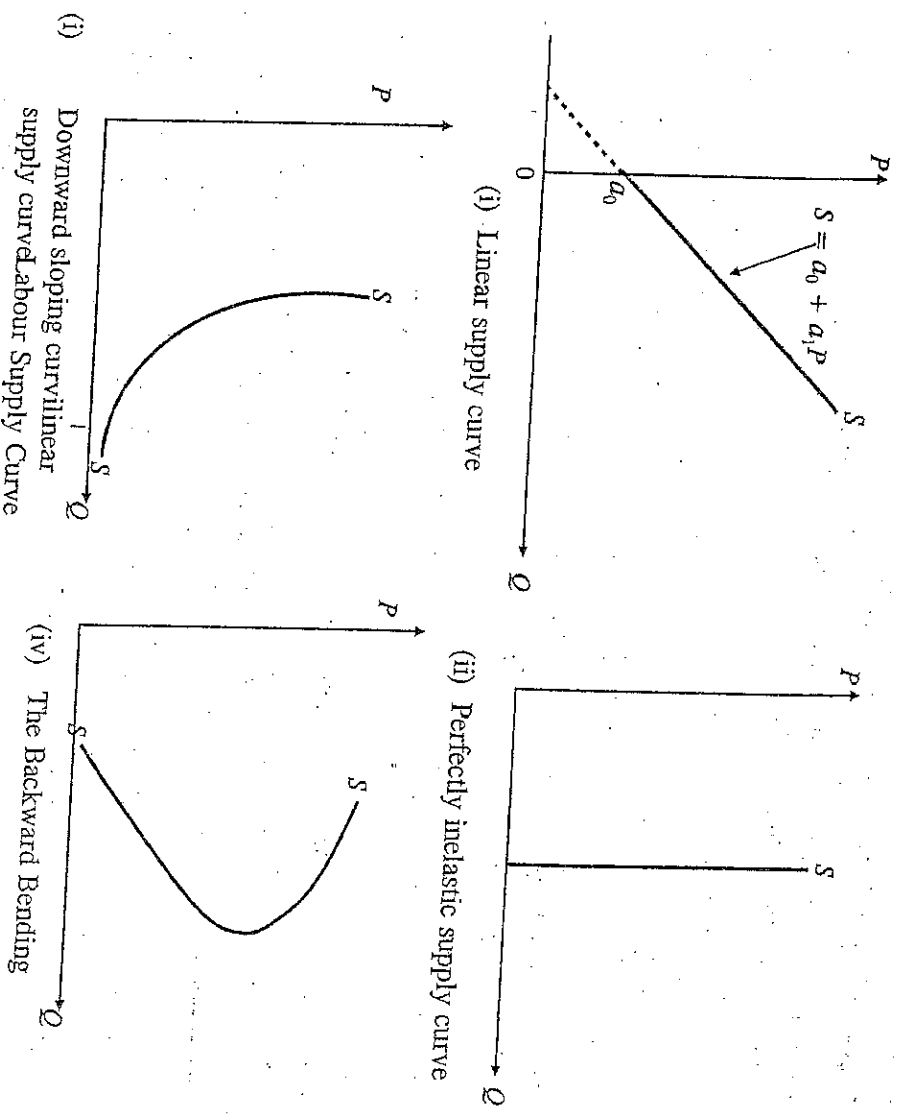


Fig. 1.3: Types of Supply Curves

SUPPLY, DEMAND AND MARKET EQUILIBRIUM

The analysis of supply and demand centres on the proposition that there exists an equilibrium price, which clears the market. An *equilibrium price* generally, refers to the attainment of a certain price from which there is no tendency to move because the economic units concerned prefer that price, namely, the buyers and the sellers in the market place. It is a state of balance between the conflicting forces operating in the laws of supply and demand.

There are two ways we can analyse equilibrium in a particular market (i) by the use of the demand and supply curves and (ii) by solving the demand and supply equations simultaneously (where they are known). We will present below the first approach while the second approach is given in the appendix to the chapter.

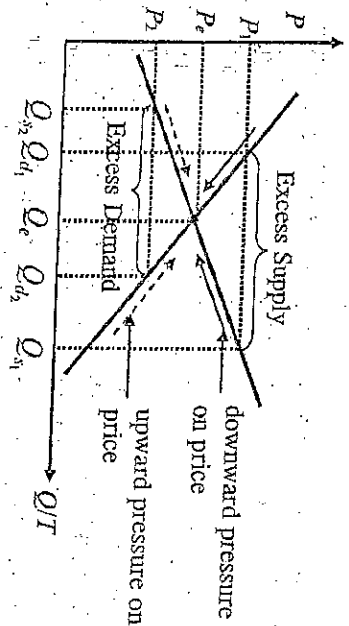


Fig. 1.4: Determination of Market Equilibrium: The Graphical Approach

The Supply and Demand Model
At price level P_1 , $P_1 > P_e$

Quantity supplied = Q_{s1} , Quantity demanded = Q_{d1}

But $Q_{s1} > Q_{d1}$ and excess supply = $Q_{s1} - Q_{d1}$

Results: downward pressure on price.

Therefore price tends to fall towards P_e

At price level P_2 , $P_2 < P_e$

Quantity supplied = Q_{s2} , Quantity demanded = Q_{d2}

But $Q_{d2} > Q_{s2}$. Excess demand = $Q_{d2} - Q_{s2}$

Results: upward pressure on prices.

Therefore prices tend to rise towards P_e .

At price level P_e , $Q_d = Q_s$, $Q_s = Q_d$

Therefore, $Q_d = Q_s$.

There is no tendency for prices to change because both consumers and suppliers have reached a preferred position on prices. This is the equilibrium price that clears the market.

The Supply and Demand Analysis we have considered above can be described as a *short-run market equilibrium*. In the short run, fixed factor inputs remain constant. Utilising more labour and raw materials can only increase output. In the long run, all factors are variable, firms have enough time to alter plant size and change other factor inputs. In addition, new firms can come into the market (therefore expanding market supply) and marginal firms can fold up (thus contracting market supply) in the long run. Thus, the *long run market equilibrium* will be different from the short run equilibrium.

Again, in our analysis, we have concentrated on a single market. This is a partial approach to the analysis of market equilibrium. *Partial equilibrium analysis* studies the

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behaviour of individual decision-making units (a household, a firm, an industry etc) and the working of individual markets in isolation. In other words, under partial equilibrium analysis, we focus on a particular economic unit or market and assume that the behaviour of other units or markets remains constant.

Much of microeconomics deals with partial equilibrium. However, the condition of a market generally affect other markets (both of factor inputs and outputs). In *General Equilibrium Analysis*, we study the behaviour of individual decision-making units and all individual markets that exists within a system simultaneously. In chapters 14 and 15, the method of analysing general equilibrium is introduced.

Finally, the study of market equilibrium can either take the form of *comparative statics* or *dynamics*. Under comparative statics, we study and compare two or more equilibrium positions, without reference to the process involved in the adjustment from one equilibrium to another. If for example, initial equilibrium price and quantity for a market is P_1 and Q_1 respectively. Assuming there is an expansion in demand over time, this will lead to a new equilibrium that may be characterised by a lower equilibrium price and larger output (P_2 and Q_2). Under comparative statics, the focus is on comparing these two equilibria without regard to the process of adjustment from one equilibrium to another. In this book, as in other intermediate micro texts, we shall be concentrating on comparative statics.

SUPPLY, DEMAND AND MARKET DISEQUILIBRIUM

Unlike demand, supply often does not respond freely to price changes. If there is an increase in the price of a product, except where there is a stock of inventory, producers must first organise the factors of production toward producing more of the commodity before there can be an increase in supply. Thus, there is a time lag in supply's response to price changes. While current demand (Q_d) is a function of the commodity's market prices (P_t), current supply (Q_s) can be regarded as a function of the current period's price (P_{t-1})

$$Q_d = f(P_t) \\ Q_s = f(P_{t-1})$$

Such relationship between market supply and demand automatically causes price to oscillate around the equilibrium level. The exact nature of the oscillation will depend on the relative strength of the price elasticities of demand (η_p) and supply (η_s). If $\eta_p > |\eta_{p-1}|$, the oscillations will be damped, decreasing in amplitude towards the equilibrium price (Fig. 1.5). If $\eta_p < |\eta_{p-1}|$, there will be explosive oscillations moving price further and further away from the equilibrium point (Fig. 1.6)

If $|\eta_p| = |\eta_{p-1}|$, the oscillations in price will be stable. In other words, price oscillates successively around the equilibrium point leading to a continuous but stable disequilibrium (Fig. 1.7).

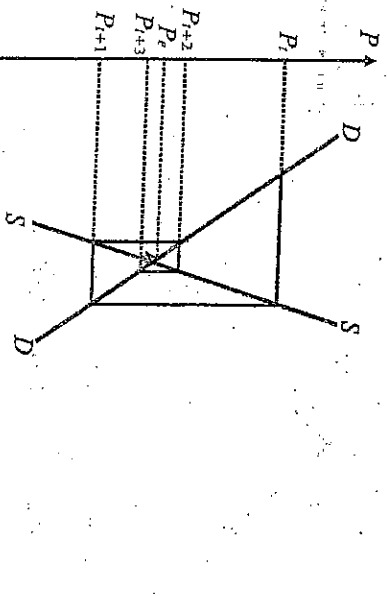


Fig. 1.5: Damped Oscillations

$$|\eta_p| > |\eta_{p-1}|$$

Price movements: $P_i \rightarrow P_{i+1} \rightarrow P_{i+2} \rightarrow P_{i+3} \dots P_e$

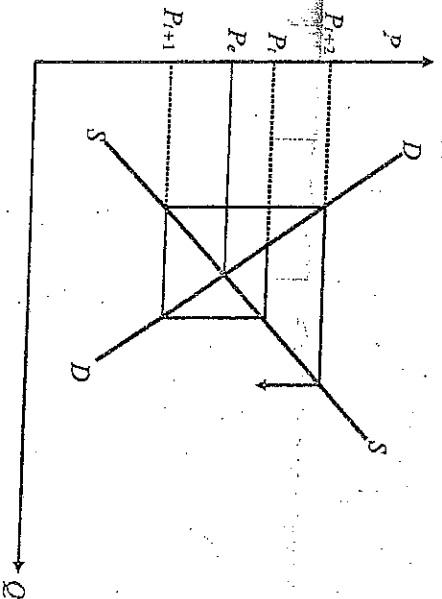


Fig. 1.6: Explosive Oscillations

$$|\eta_p| < |\eta_{p-1}|$$

Price movements: $P_i \rightarrow P_{i+1} \rightarrow P_{i+2} \dots P_{i+n}$

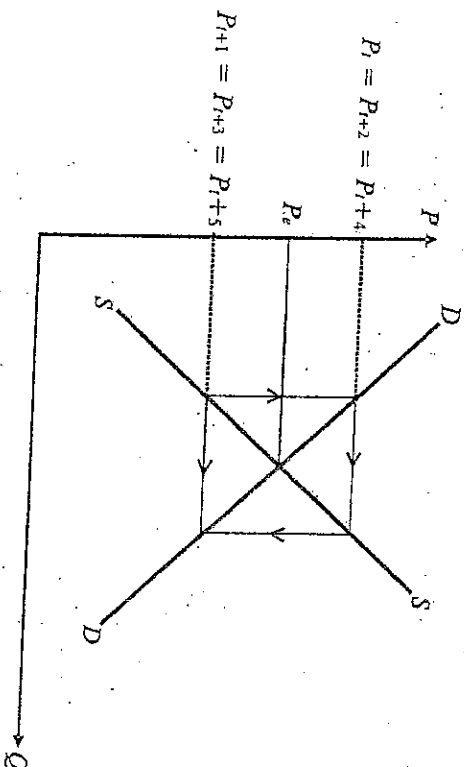


Fig. 1.7: Stable Oscillations

$$|\eta_p| = |\eta_{p^{t+1}}|$$

Price movements: $P_t \rightarrow P_{t+1} \rightarrow P_{t+2} \rightarrow P_{t+3} \dots P_{t+n}$

Such market's situation analysed above (the cobweb model) is often associated with perishable products where it is not possible to keep inventories, particularly agricultural products.

The market for abnormal (ostentatious) and subnormal (Giffen) goods may also exhibit a special type of disequilibrium, which though not oscillatory, is usually explosive. This is because, in both instances, we have a situation of an upward sloping demand curve combining with an upward sloping supply curve.

In Fig. 1.8, assuming initial price and quantity equilibrium is at P_e and Q_e respectively, if we take the case of an ostentatious good and assume that price increases from P_e to P_{t_1} . The rise in price will cause the demand to increase to Q_{t_1} while supply is Q_{t_1} . The excess demand $Q_{t_1}Q_{t_1}$ will push price further upward to P_{t_2} .

The reverse will be the case if we assume a price fall, supply exceeds demand and the excess supply causes price to fall further.

If we take the case of a Giffen good. Assume initial equilibrium is at P_e , Q_e assuming further that price falls to P_{t_1} . At P_{t_1} , supply exceeds demand by Q_{t_1} , Q_{t_1} . This forces price to fall further to P_{t_2} .

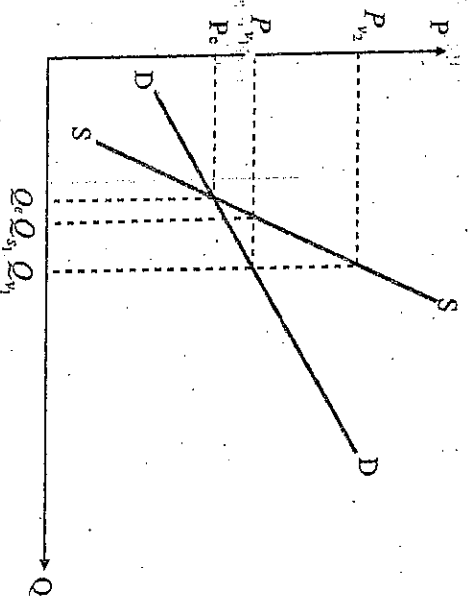


Fig. 1.8: One-way explosive Disequilibrium: The Case of an Ostentatious good.

From the above analysis, it is obvious that equilibrium can be rarely obtained in the markets for agricultural goods whose supply exhibits the characteristic defined by equation 1.13, and in the markets for abnormal and subnormal goods. If by chance equilibrium is attained, such equilibrium cannot be a stable one. Equilibrium in such markets is balanced on a knife-edge, and any small change in price will send the system into a vicious disequilibrium.

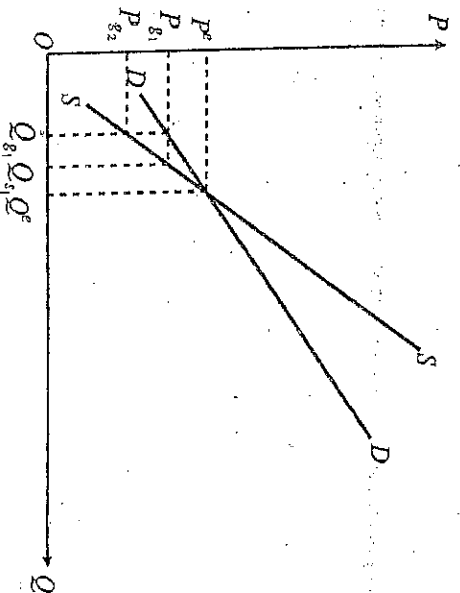


Fig. 1.9 One-way explosive disequilibrium: The case of a Giffen good.

MARKET FAILURE

Microeconomic theory is founded on the premise of a smooth functioning market. The market (price) system allocates resources by providing information to resource owners on the relative price of individual goods and services. The price mechanism is a system whereby the information provided by the relative prices of goods and services determines freely what the society is to produce, how and for whom it is to be produced. The price system is assumed to allocate resources effectively, bringing maximum benefits to the society and resource owners.

There are three main arguments proffered in support of the price system as a means of resource allocation.

- (i) its ability to achieve optimality in the allocation of societal resources
- (ii) its built-in flexibility which negates the need for conscious government planning and intervention, and
- (iii) its guarantee of economic freedom to all and sundry.

In spite of its benefits, the market mechanism is not without its limitations.

"Market failure" is a phrase that is usually employed to describe one or both of two things

- (i) the failure of the price mechanism, at any point in time, to achieve optimal resource allocation
- (ii) the inability of the price system to achieve other desirable societal goals associated with the distribution and utilisation of her resources.

Some of these goals may be economic (e.g., achieving an equitable distribution of society's income) or non-economic (such as preserving society's value system). It has been asserted that the greed and selfish interest of individuals in the society, nurtured by the price system, may lead to actions, which enriches a few at the expense of the society at large.

In relation to optimal resource allocation and utilisation, the price mechanism does not, by itself, accommodate externalities in economic transactions. Again, the system is insensitive to the provision of public goods. In the case of the "commons" (common property resources), the price system offers no protection. It does not tell us how such resources are to be prevented from over-exploitation or how they are to be allocated in a way that the larger society is benefited.

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Finally, the price mechanism breaks down whenever and wherever there are inefficient transportation and communication systems. With market imperfections, there is bound to be resource unemployment and imbalances within the economy.

Market Failure and Microeconomic Policies

Market failures are often corrected through appropriate microeconomic policies. A microeconomic policy is a deliberate policy of government (or its agencies) aimed at influencing the allocation of resources (production and distribution) along some desired lines. There are three main tools of microeconomic policies

- (1) Rule Making which calls for direct government intervention in the workings of the free market system. Examples are the use of quota restriction on the importation of certain items, banning the production and sale of a particular commodity etc.
- (2) Transfer Payments and Public provisions of goods and services. Usually, the objective of this is to redistribute societal income in favour of the less privileged.
- (3) Structural incentives aimed at altering market signals with a view to influencing resource allocation along desirable lines. Examples are the use of subsidies, imposition of taxes on the production and sale of a particular item, the use of administered prices (price controls), scholarships etc. Often, structural incentives are employed by Government to correct for externalities in production and distribution.

In the next section, we shall examine the use of administered prices as a tool of microeconomic policy.

ADMINISTERED PRICES

Administered prices are prices that are fixed by the government, designed to hold the selling price of a commodity or service at a disequilibrium value. It is directed at correcting excessively high prices of some goods and excessively low prices of some factor inputs. Administered prices are called controlled prices and are usually administered through the legal process by instituting *price ceilings* and *price floors*.

Price Ceilings

A price ceiling is a legally established *maximum price*. Because a price ceiling is usually imposed below the equilibrium price, consumers will always want to buy more

at this price than producers are willing to supply. This results in shortages in the market (represented by ab in Figure 1.10).

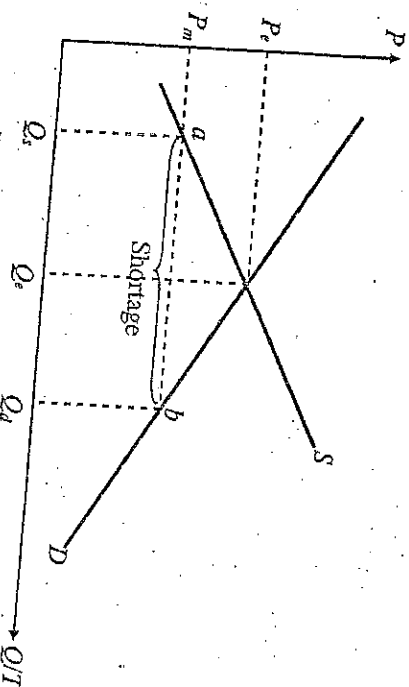


Fig. 1.10: Impact of a Price Ceiling

Market equilibrium price is P_e and equilibrium quantity is Q_e . Price ceiling is fixed at P_m . At this ceiling price, demand (Q_d) exceeds supply (Q_s) leading to a market shortage of $Q_dQ_s = ab$.

Price ceiling always causes a disruption in the market clearing process. The maintenance of a price ceiling requires that the government must replace the market mechanism with some other way to allocate goods. One way of doing this is *rationing*. In many instances where price ceilings are imposed, black markets always emerge. *Black markets* are market processes in which participants buy and sell the goods that are controlled by price ceilings at prices above the legally established one.

For a price ceiling to be effective as a means of resource re-allocation, the government must be able to prevent the occurrence or eliminate the existence of black markets.

This can only be achieved when the government is willing and able to meet the shortfall in supply, either through direct provision of the good concerned, by importation, or by granting subsidies to firms engaged in the production of such commodity.

Price Floors

Price floors are legally imposed *minimum prices* usually above the market equilibrium price. Price floors are usually encountered in the case of the pricing of factor inputs, especially labour. Minimum wage legislation is common in all societies.

Price floors usually result in market surplus, because supply will always exceed demand at the floor price. (Figure 1.11). For price floors to be successful therefore, the government must find a way to accommodate the surplus. In the case of commodities, the government may decide to buy over the surplus goods. In the case of factor inputs,

like labour, the government would have to create ways to employ the surplus labour directly or by taking on policies that will expand the productive base of the economy and her capacity to employ labour resources. The latter step is, however, a long run situation and would not meet the immediate need of what to do with the surplus created by the minimum price.

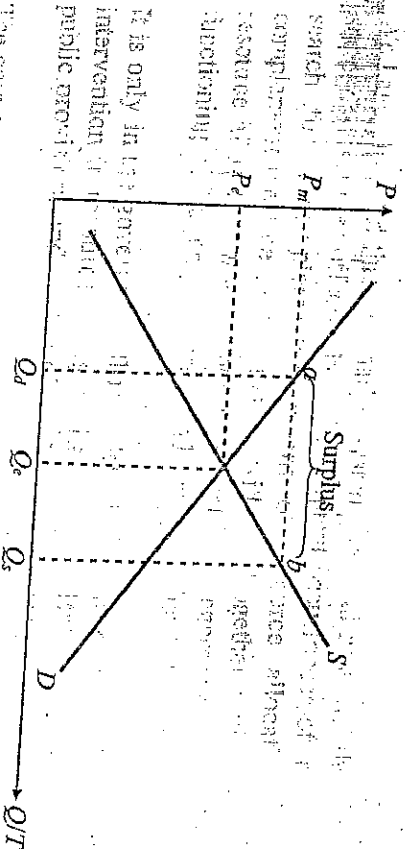


Fig. 1.11:

At equilibrium price P_e , $Q_d = Q_s = Q_e$. With price floor at P_m above P_e , $Q_s > Q_d$ and a surplus of $Q_s - Q_d = ab$ results.

COSTS AND BENEFITS OF GOVERNMENT INTERVENTION IN RESOURCE ALLOCATION

We have observed that the market system does fail now and then in its effort to allocate society's resources effectively. The frequent occurrences of market failures lead to the search for a complementary means of resource allocation, a method that will complement the free market system without altogether replacing it. Such a means of resource allocation would have the objective of encouraging and enforcing the smooth functioning of the market system and correcting market failures whenever it occurs.

It is only in the government that society finds this complement. However, government intervention in resource allocation has its cost. Rule making is expensive, so also are public provisions and the use of administered prices.

The cost of government intervention in resource allocation is both internal and external. Costs internal to the government are incurred directly by government itself. They include the cost of processing, formulating and administering rules and the cost of prosecuting offenders, the cost of directly providing for a commodity in short supply in the event of a price ceiling, and the cost of having to buy off excess supply in the case

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of price floors. Direct external cost to the government include the additional costs incurred by firms in keeping with government regulations, while indirect external costs are costs to the society as a result of delays created by government regulations.

In addition to the cost, government intervention in resource allocation does not at all times achieve its objective. Like the market system, government allocation is not immune from failure. The ability of the government to efficiently intervene in resource allocation will vary with societies. However, in general, certain factors militate against effective government intervention in resource distribution. These include:

- (i) political factors which may prevent the adoption of the right policy at the right time
- (ii) the corrupt and unreliable nature of the civil service in many societies which hinders the smooth administration of government's policies
- (iii) the absence, in most societies, of a dynamic system which ensures that government's rules and regulations change with changing economic and technical conditions in the real world and
- (iv) the absence of perfect knowledge on the part of government.
(Lipsey *et al.*, 1998: 424-425, also in Iyoha *et al.*, 1998: 424-425).

Thus, it is imperative that government must weigh the benefit of any action taken along the line of influencing resources distribution against the cost. Where the cost outweighs the benefits, the market is best left alone. The general rule is for government to concentrate on those areas where market allocation has been known to be grossly inefficient (such as correcting for externalities, protecting the commons and providing public and social goods which are unattractive to the private sector), while leaving the market system to answer the other questions of resource allocation.

APPENDIX

DETERMINATION OF MARKET EQUILIBRIUM: THE QUANTITATIVE APPROACH

1. The case of a single Product (X)

Given a simple linear Demand and Supply function,

$$Q_d = 6 - 0.5P$$

$$Q_s = -4 + 1.5P$$

(1)

(2)

At equilibrium, $Q_d = Q_s = Q_e$

Equating (1) in (2),

$$6 - 0.5P = 4 + 1.5P$$

$$P = 5$$

Substituting value of P into (1) or (2)

$$Q_e = 3.5$$

Thus, equilibrium price and quantity are $P_e = 5$ and $Q_e = 3.5$

2. The Case of Two Related Goods (X and Y)

Given two substitute goods (X and Y) with Demand and Supply functions,

$$Q_d = 41 - 1.5P_x + 0.5P_y \quad (3)$$

$$Q_s = -2.5 + 7.5P_x \quad (4)$$

$$Q_d = 46 + P_x - 2P_y \quad (5)$$

$$Q_s = -3 + 16P_y \quad (6)$$

At equilibrium,

$$Q_d = Q_s = Q_e$$

$$Q_d = Q_s = Q_e$$

Equating (3) and (4)

$$41 - 1.5P_x + 0.5P_y = -2.5 + 7.5P_x$$

$$9P_x - 0.5P_y = 43.5 \quad (7)$$

Equating (5) and (6)

$$46 + P_x - 2P_y = -3 + 16P_y$$

$$-P_x + 18P_y = 49 \quad (8)$$

Equations (7) and (8) give us a simultaneous equation system. Solving simultaneously

$$P_x = 5 \text{ and } P_y = 3$$

Substituting value of P_x into (3) or (4),

$$Q_e = 35$$

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Substituting value of P_y into (5) or (6)

$$Q_{xy} = 45$$

Thus, equilibrium price and quantity for each product is

$$P_x = 5, \quad Q_{ax} = 35$$

$$P_y = 3, \quad Q_{xy} = 45$$

3. The Case of Three or More Related Goods

Where there are more than two goods, the solution for equilibrium output and quantity for each good can be obtained by using the same method employed above. Where there are three related goods (X, Y, Z), the solution will yield a simultaneous equation system with three equations and three unknowns, P_x , P_y and P_z . The solution can be solved by substitution (or elimination) or by matrix algebra.

4. The Case of Curvilinear Demand and Supply Functions

Given the following demand and supply functions for a commodity (X),

$$2P = -2Q^2 - 6Q + 40 \quad (9)$$

$$2P = 6Q^2 - 20Q + 10 \quad (10)$$

Rearranging

$$2P + 2Q^2 + 6Q - 40 = 0 \quad (9)$$

$$2P - 6Q^2 + 20Q - 10 = 0 \quad (10)$$

At equilibrium, $Q_{dx} = Q_{sx}$,

$$2P + 2Q^2 + 6Q - 40 = 2P - 6Q^2 + 20Q - 10 \quad (11)$$

$$8Q^2 - 14Q - 30 = 0$$

Since Q cannot be negative, $Q = 3$, or $Q = -1.25$

We can substitute the value of Q into (9) or (10) to get P .

Thus equilibrium price and quantity is $P = 2$, $Q = 3$.

5. Deriving a Demand or Supply Function from a Schedule

When a demand or supply schedule is linear, it is always possible to derive the function from the schedule.

Given the demand schedule

P	Q_d
1	35
2	30
3	25
4	20
5	15
6	10

The first step is to note the successive change in Q_d for every one unit change in P . If this is constant, then the schedule is linear and we can derive the function. Since the general form of a linear demand function is $Q_d = a + bP$, we only need to find the two constants, a and b . The constant difference in $Q_d(\Delta Q_d)$ as a result of a one unit change in $P(\Delta P)$ will form the coefficient of P in the function. It also constitutes the slope of the function.

In our example $\frac{\Delta Q_d}{\Delta P} = -5$. For every one-unit increase in P , Q_d falls by 5 units.

Thus $b = -5$.

The next step is to find the autonomous term (a). To do this, multiply the absolute value of the constant difference ($-b$) by the first value of P and add to the first value of Q_d .

$$|-5| (1) + 35 = 40.$$

$$a = 40$$

Thus, the function can be expressed as

$$Q_d = 40 - 5P$$

the result can be verified by attempting to derive the schedule from the function.

Questions for Discussion

1. What are the central issues in the study of microeconomics? Explain the assumption and postulations of classical microeconomics.
2. Examine the validity of the law of Demand and supply.
3. (a) What do we mean by "market equilibrium"? Differentiate between

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- (i) static and dynamic equilibrium analysis
 - (ii) partial and general equilibrium analysis
- (b) Explain the concept of a "knife-edge equilibrium". How relevant is this concept to the market for "second-hand clothing" or the market for "ice fish" in Nigeria?
4. What are the basic assumptions of the Cobweb model? How relevant is the analysis to pricing, supply and demand behaviour in the real world? (Iyoha et al, 1998: 227).
5. (a) What is market failure? How can such failure be corrected?
(b) What are the costs of government intervention in resource allocation?
6. What are administered prices? With the aid of well-labelled diagrams, explain the benefits and problems associated with the use of such prices in allocating resources.

Problems

1. Given the demand and supply functions

$$Q_d = 25 - 5P$$

$$Q_s = -15 + 2.5P$$

determine the equilibrium price and quantity, and illustrate your answer graphically.

2. Find the equilibrium price and quantity for two complementary goods, X and Y, whose demand and supply functions are given as

$$Q_d = 205 - 2.5P_x - P_y$$

$$Q_s = -30 + 1.5P_x$$

$$Q_d = 147.5 - 0.5P_x - 1.5P_y$$

$$Q_s = -60 + P_y$$

3. Given the demand and supply equations for three complementary goods (X, Y and Z),

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$$Q_x = 11.5 - 2.5P_x - 0.5P_y - 0.5P_z$$

$$Q_x = -4 + 3P_x$$

$$Q_x = 7.5 - 0.5P_x - 1.5P_y - P_z$$

$$Q_{xy} = -5.5 + 1.5P_y$$

$$Q_x = 9.5 - 0.5P_x - P_y - 2P_z$$

$$Q_x = -2.5 + 1.5P_z$$

Find equilibrium price and quantity in each market.

2. Find the equilibrium price and quantity from the following demand and supply functions

$$4P = -4Q^2 - 12Q + 120$$

$$4P = 12Q^2 - 40Q + 30$$

3. You are given the following data on the demand and supply of commodity X

Price (₹)	Quantity demanded
5	0
4	60
3	120
2	180
1	240
0	300

Price (₹)	Quantity supplied
2	0
3	30
4	60
5	90
6	120
7	150

Expressing price as a function of quantity, you are required to

- derive the demand function
- derive the supply function
- solve for the equilibrium quantity exchanged in the market and the equilibrium price using (i) and (ii) above.
- illustrate your answer in (i) to (iii) graphically.

4. Given the following functional relationships

$$Q_d = f(P)$$

$$Q_s = f(P_{-1})$$

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and the hypothetical demand and supply schedules for two commodities, X and Y, as follows

P_i	$P_{L,i}$	Q_d^x	Q_s^y	Q_d^x	Q_s^y
1	-	60	10	120	40
2	1	55	15	80	60
3	2	45	21	50	100
4	3	35	28	30	150
5	4	15	32	10	200

What do you think is the nature of the disequilibrium that may exist in the market for each good? Justify your answers.

References

- Fusfield, Daniel R. *The Age of the Economist*. Scott Foresman and Company, 1972. Chapter 3.
- Iyoha M.A, Oyefusi, S.A and Oriakiti, D.E (1998), *An Introduction to Modern Economics* March Publishers, Benin City, Nigeria. Pp. 424-425.
- Lipsey, R.G, Steiner, P.O and Purvis, D.D 1971), *Economics*, Harper and Row Publishers, New York. Pp. 421-425.
- Smith Adam (1982), *The Wealth of nations*, Pegium Books Ltd., England.

depends upon their costs of production and the price they expect to receive per unit of output.

If there are n firms in the market, market supply is the horizontal summation of the individual firm's supplies. This can be expressed symbolically as,

$$S_m = \sum_{i=1}^n P q_i \quad (1.6)$$

Where P is the market price per unit and q_i is the quantity supplied by the i th firm.

The Supply Function

The quantity of goods that firm's will provide depends on some key variables namely, the state of technology, prices of inputs, prices of other goods closely related to the commodity in production, future price expectations, the firm's goals and weather, strikes and other short-run forces.

All these variables play a role in what is called the supply function. More specifically, the supply function is a statement specifying how each of a number of relevant variables influences the quantity of a product that firms will provide at any given period of time.

The supply function can be expressed symbolically as,

$$Q_s = [P, P_n, E, T, O, M] \quad (1.7)$$

Where P = Price of the product

P_n = Price of the related goods: complements and substitutes in production

E = Expectation about future prices of goods

T = State of technology

O = Other short run factors

M = Number of suppliers

The Supply Schedule and the Law of Supply

The supply schedule is a table showing the relationship between various possible prices of a product and the quantity provided per unit of time, other things being equal. This relationship is direct (positive) for a normal good, hence, the supply curve will normally be upward sloping. The direct relationship between the quantity supplied and the price level is known as the law of supply. More generally, the law of supply states that over a specified period of time, firms will tend to offer for sale larger quantities of a commodity when the price is high than when the price is low. This law can be expressed symbolically as

$$P \uparrow \rightarrow Q_s \uparrow \text{ and } P \downarrow \rightarrow Q_s \downarrow$$